AN R-PARTITE ORDERED HYPERGRAPH MATCHING REALISING PRESCRIBED CLIQUE-SIZE BOUNDS VIA A MONOTONE POINT SET

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Allowed standard prerequisites (editorial).

- The multi-dimensional Erdos-Szekeres theorem in its second form, producing a point set with bounded monotone subsequences. The author's theorem labelled thm_es (source lines 232-238), which the counted proof invokes by label. Produces the point collection Z whose monotone sequences have the prescribed bounded lengths.
- The correspondence between r-patterns of the finest ordered partition and monotone functions, and the quantity L_P of a collection. Established at the end of the author's preliminaries (discussion at the end of the section, source lines 281-288) and used in the counted proof. Turns the bounded monotone sequences of the point set into the clique-size bounds L_P of the constructed matching.

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Source reference. Original section 2: sources/combinatorics-fl02-a-sharp-ramsey-theorem-for-ordered/pinned-original-source.tex, source lines 226–226. Navigation pointer only: it locates the author's preliminaries where the pattern parameterisation is set up; no text is reproduced.

Source reference. Original theorem 2.1: sources/combinatorics-fl02-a-sharp-ramsey-theorem-for-ordered/pinned-original-source.tex, source lines 232–232. Navigation pointer only: the counted proof applies this multi-dimensional Erdos-Szekeres theorem; it is declared as a named standard prerequisite and is not reproduced.

Source reference. Original theorem 1.2: sources/combinatorics-fl02-a-sharp-ramsey-theorem-for-ordered/pinned-original-source.tex, source lines 199–199. Navigation pointer only: the counted proof refers to the author's upper-bound theorem in the surrounding discussion; the theorem is not reproduced.

Source reference. Original theorem 4.3: sources/combinatorics-fl02-a-sharp-ramsey-theorem-for-ordered/pinned-original-source.tex, source lines 455–455. Navigation pointer only: the counted construction is the building block of that theorem, which has its own card in this lane.

Source reference. Original equation 4.1: sources/combinatorics-fl02-a-sharp-ramsey-theorem-for-ordered/pinned-original-source.tex, source lines 549–549. Navigation pointer only: it labels the author's display defining L_P, used by the counted proof; the display is not reproduced.

Lemma 0.1. *Let r be a positive integer, and let $\lambda^{(r)} = (1, \dots, 1)$ be the unique ordered partition of r into r parts. For every r-pattern $P \in \mathcal{P}(\lambda^{(r)})$, consider a positive integer m_P. Then there exists an r-partite ordered r-uniform hypergraph matching $\mathcal{H}$ of size*

$$|\mathcal{H}| = \prod_{P \in \mathcal{P}(\lambda^{(r)})} m_P,$$

such that $L_P(\mathcal{H}) \leq m_P$ for all r-patterns $P \in \mathcal{P}(\lambda^{(r)})$ and $L_P(\mathcal{H}) = 1$ for all other r-patterns $P \notin \mathcal{P}(\lambda^{(r)})$.

Proof of Lemma 0.1. Recall from the discussion at the end of Section 2 that any r-pattern $P \in \mathcal{P}(\lambda^{(r)})$ is of the form $P = P(\lambda^{(r)}, \tau)$ for a unique function $\tau : \{1, \dots, r\} \to \{1, -1\}$ with $\tau(1) = 1$. For every such function τ, let us define $m_\tau = m_{P(\lambda^{(r)}, \tau)}$. By the second part of Theorem 2.1, there exists a collection of points $Z \subseteq \mathbb{R}^r$ of size $|Z| = \prod_\tau m_\tau = \prod_{P \in \mathcal{P}(\lambda^{(r)})} m_P$ such that the $|Z| \cdot r$ coordinates of all the points in Z are all distinct, and such that for every function $\tau : \{1, \dots, r\} \to \{1, -1\}$ with $\tau(1) = 1$, every τ-monotone sequence in Z has length at most $m_\tau = m_{P(\lambda^{(r)}, \tau)}$.

Without loss of generality, we may assume that all points in Z have positive coordinates (otherwise, we can translate the set Z accordingly). Let $N > 0$ be a number larger than all coordinates of all points in Z. Now, for every point $z = (z_1, \dots, z_r) \in Z \subseteq \mathbb{R}^r$, let us define $e(z) = \{N + z_1, 2N + z_2, \dots, rN + z_r\} \subseteq \mathbb{R}$, and note that the sets $e(z)$ for $z \in Z$ are pairwise disjoint. Taking $\mathcal{H} = \{e(z) \mid z \in Z\}$ to be the collection of the sets $e(z)$ for all $z \in Z$, we obtain an ordered r-uniform hypergraph matching $\mathcal{H}$ by considering the natural ordering on $\mathbb{R}$ on the vertex set. Note that $\mathcal{H}$ has size $|\mathcal{H}| = |Z| = \prod_{P \in \mathcal{P}(\lambda^{(r)})} m_P$. Furthermore, $\mathcal{H}$ is r-partite (by the choice of N) and so in particular we have $L_P(\mathcal{H}) = 1$ for any $P \notin \mathcal{P}(\lambda^{(r)})$. It remains to check $L_P(\mathcal{H}) \leq m_P$ for every $P \in \mathcal{P}(\lambda^{(r)})$.

So let $P \in \mathcal{P}(\lambda^{(r)})$ and let $\tau : \{1, \dots, r\} \to \{1, -1\}$ with $\tau(1) = 1$ be such that $P = P(\lambda^{(r)}, \tau)$, then we have $m_\tau = m_P$. Let $z^{(1)}, \dots, z^{(t)} \in Z$ be such that the edges $e(z^{(1)}), \dots, e(z^{(t)})$ form P-clique in $\mathcal{H}$. Then for any $1 \leq \ell < \ell' \leq t$,

for any $i \in \{1,\dots,r\}$ with $\tau(i) = 1$, the i-th vertex $iN + z_i^{(\ell)}$ of $e(z^{(\ell)})$ comes before the i-th vertex $iN + z_i^{(\ell')}$ of $e(z^{(\ell')})$ and so we have $z_i^{(\ell)} < z_i^{(\ell')}$. Similarly for any $i \in \{1,\dots,r\}$ with $\tau(i) = 1$, the i-th vertex of $e(z^{(\ell)})$ comes after the i-th vertex of $e(z^{(\ell')})$ and so we have $z_i^{(\ell)} > z_i^{(\ell')}$. Thus, for every $i \in \{1,\dots,r\}$ with $\tau(i) = 1$, the i-th coordinates of the points $z^{(1)},\dots,z^{(t)}$ form an increasing sequence $z_1^{(1)} < \cdots < z_i^{(t)}$, and for every $i \in \{1,\dots,r\}$ with $\tau(i) = -1$, they form a decreasing sequence $z_1^{(1)} > \cdots > z_i^{(t)}$. Thus, $z^{(1)},\dots,z^{(t)}$ is a τ-monotone sequence of points in Z and hence $t \le m_\tau \le m_P$. This shows that $L_P(\mathcal{H}) \le m_P$, as desired. □

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